D1 — Project Proposal and Charter

Boréal Marché Customer Intelligence Platform · Capstone Project 1

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1. Problem statement — in the client's terms

Boréal Marché (fictional Québec online grocery and specialty retailer; ~5,000 registered customers, ~4,000 active SKUs, ~25,000 orders a year across Québec, Ontario and the Maritimes) runs three parts of its business on intuition. Each is a separate, measurable problem:

#	Today	Why it costs money	What we will measure
1	**Retention.** A 15% coupon goes to *every* customer *every* quarter.	Margin is given away on customers who would have ordered anyway, while customers who are quietly lapsing get the same untargeted offer.	Campaign profit per quarter vs. "coupon to everyone" (CAD), using stated cost assumptions
2	**Cross-sell.** The "you might also like" strip was hard-coded by an intern eight months ago.	It reflects no purchasing behaviour; every basket is a missed attach opportunity.	Hit-rate@5 of recommendations on future baskets vs. a bestseller list
3	**Customer support.** Two employees answer the same ~40 questions (delivery, returns, allergens, privacy) all day, in two languages.	Two people on repetitive work; slow answers; inconsistent policy statements.	Share of golden-set questions answered correctly and grounded; 100% refusal on out-of-scope

The platform answers each with one *head* on one shared data pipeline: **A — Prédiction** (will this customer order again in the next 90 days?), **B — Recommandation** (given this basket, what next?), **C — Assistant** (grounded bilingual answers that call A and B through our own API).

2. Dataset selection and justification

Choice: UCI Online Retail II (Chen, 2012; DOI 10.24432/C5CG6D; licence **CC BY 4.0**, which permits educational use and redistribution with attribution). One workbook, two yearly sheets, 1,067,371 transaction lines, 2009-12-01 → 2011-12-09, 53,628 invoices, 5,305 stock codes, 5,942 customer ids. File SHA-256 `bcbe73b35f5b…` (full value in `data/manifest.json`), 43.5 MB.

Required property	Present?	Evidence
Multi-item transactions	Yes	25.4 lines per clean invoice on average → association rules are meaningful
Timestamps	Yes	Minute-level `InvoiceDate` over 2 years → observation/label windows and a temporal holdout
Customer identifier	Yes, partially	`Customer ID` present on 776,577 of 1,003,338 clean lines; guests (22.6%) are unusable for Head A but valid baskets for Head B

Known quality issues (first EDA, all handled by documented rules in D2/D3): cancellations (invoices prefixed `C`, negative quantities), non-positive prices, service/fee codes (`POST`, `DOT`, `M`, `BANK CHARGES`, `AMAZONFEE`, test and gift-voucher codes), the same SKU under several descriptions, extreme bulk lines (e.g. an 80,995-unit order cancelled the same day), missing customer ids, and — found during this proposal — **the two yearly sheets overlap for 2010-12-01…09**, so naive concatenation double-counts 22,523 lines.

Honest provenance. The data are a UK giftware wholesaler's real transactions, *re-skinned* as Boréal Marché for a pedagogical scenario: countries map to delivery regions (UK → QC, continental Europe → ON, Ireland/islands → MARITIMES, rest → INTL), a keyword category is added, amounts are labelled CAD without conversion. The mapping is in `DATA_CARD.md` and `src/boreal_marche/reskin.py`. This is not proprietary client data and is never presented as such.

Consequences we accept: the catalogue is giftware, not groceries; many customers are small businesses (large baskets). The raw file is DVC-tracked and never committed to Git; one command (`make data`) fetches it, verifies the checksum and writes the manifest. Alternatives (Instacart, Olist, Dunnhumby) were not needed: Online Retail II satisfies all three criteria and is the course default.

3. Proposed architecture — offline and online halves

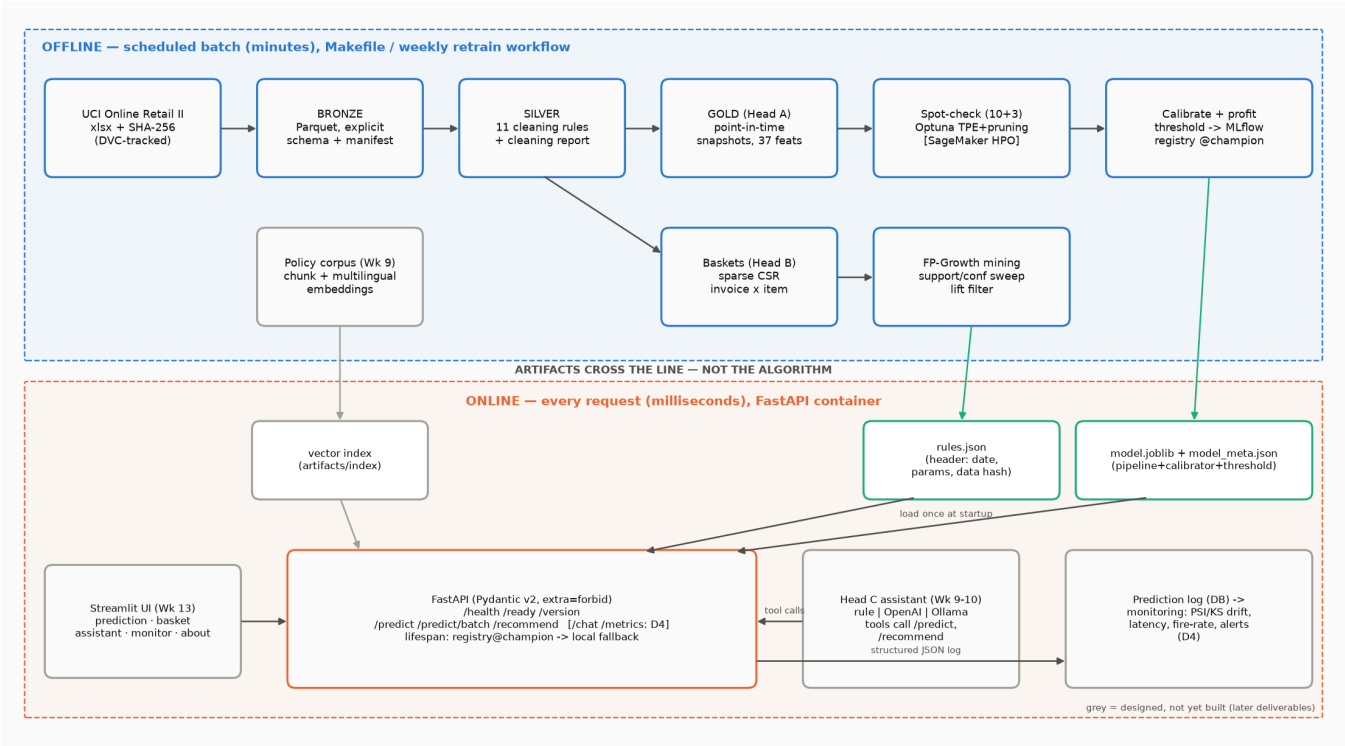


Figure 1 — Offline half (blue) runs on a schedule and produces artifacts; online half (orange) only loads artifacts and answers requests. Grey components are designed now and built in later weeks.

Mining and training are expensive and run offline; serving is a lookup. Only artifacts cross the line: `model.joblib` + `model_meta.json` (Head A), `rules.json` (Head B), a vector index (Head C). One FastAPI service exposes `/predict`, `/recommend` and `/chat`; one Streamlit interface and one monitoring dashboard consume it.

4. Success metrics — targets fixed in advance

Head	Metric (primary in bold)	Baseline it must beat	Target	Business translation
A	**PR-AUC on the held-out test snapshot**	Dummy prior; one-rule "ordered in last 60 days"	≥ 0.62 and ≥ +0.05 over the rule	—
A	Brier score / reliability after calibration	Uncalibrated model	ECE ≤ 0.05	Probabilities usable in a dollar calculation
A	Campaign profit at the chosen threshold	Coupon to everyone (status quo) and to nobody	> status quo, in CAD per quarter	Margin recovered
B	**Hit-rate@5** on a temporal holdout	Top-5 bestsellers	≥ 2× popularity; coverage ≥ 40% of baskets	More relevant attach items
C	Tool-choice accuracy on a 40-question golden set (in CI)	Rule/keyword provider	≥ 85%; 100% refusal on out-of-scope	Support workload deflected
Eng.	`/predict` p95 latency · image size · CI runtime	—	< 150 ms · < 1 GB · < 8 min	Cheap to run
Cost	Total cloud spend	—	< $12 (hard ceiling $25)	—

If a head does not beat its baseline, that is reported as a finding, not hidden.

5. Scope

In scope: the three heads, reproducible ETL, experiment tracking and registry, typed API, containers, CI/CD, a deployed interface and monitoring, Law 25 governance artifacts. **Out of scope, and why:** real-time streaming ingestion (batch is enough for quarterly campaigns); causal uplift modelling (no randomised coupon data exists — we model *propensity* and state the limitation); collaborative filtering and deep recommenders (no ratings, sparse repeat purchase — see D2); fine-tuning an LLM (RAG is cheaper and auditable); a mobile app; any real customer data or contact.

6. Risk register

#	Risk	Likelihood	Impact	Mitigation	Owner
R1	**Data quality** — duplicates, cancellations, overlap between sheets distort features	High (confirmed)	High	Ordered cleaning rules with row counts, data-quality test suite in CI	Olga
R2	**Leakage** — features computed across the label window give fake 0.95+ scores	Medium	High	Point-in-time feature code with unit tests; deliberate leaky experiment; test set touched once	Olga
R3	**Cloud cost** — forgotten SageMaker endpoint or App Runner service	Medium	Medium	$10 budget alarm (Week 1), managed spot, `max_run` caps, teardown checklist, Render free tier for serving	Olga
R4	**Scope creep** — three heads + full MLOps for one person	High	High	Skeleton first (stub API deployed by Week 5), STRETCH items only after D4; weekly scope check	Olga
R5	Seasonality/drift — Q4 peak changes base rates between train and test	High	Medium	Quarterly cut-offs covering all seasons; calibration on a separate snapshot; report shift explicitly	Olga
R6	Hosted-LLM privacy — personal data sent outside Québec	Low (design)	High	No personal data to hosted providers; local Ollama provider; Law 25 s. 17 review	Olga
R7	Environment drift — host Python 3.14 vs. required 3.11; library upgrades	Medium (confirmed)	Medium	`uv`-managed Python 3.11, pinned ranges, Docker image, CI on 3.11	Olga
R8	Single point of failure — solo student illness/overload	Medium	High	14 h/week budget with buffer weeks 7 and 13; early submissions of skeletons	Olga

7. Team charter (individual project)

- **Role:** Olga Morgunova owns data, modelling, serving, operations and documentation.
- **Weekly time budget:** 14 h (2 h studio + 12 h project/homework; 2-11-5 ponderation = 210 h over 15 weeks). Split ≈ 55% build, 20% evaluation, 25% documentation.
- **Working agreement:** work on short-lived branches, pull requests into `master` with green CI; self-review with a checklist (tests, numbers traceable to `reports/metrics`, no secrets) and a 24-hour cooling-off before merging report text; AI-assistant output is reviewed line by line.
- **Cadence:** Monday plan (issues/milestones), Thursday studio checkpoint with the instructor, Sunday retrospective in `PLAN.md` (decision log).
- **Review:** peer review of the feature table (Week 3) and cold-start test by a classmate (Week 11).

8. Fifteen-week plan (mapped to the course milestones)

Wk	Build task	Checkpoint / due
1	Repository, branch protection, tooling, budget alarm	Repository review

Wk	Build task	Checkpoint / due
2	Ingestion (`make data`), cleaning report, **this proposal**	**D1 (10%)**
3	Point-in-time feature table + tests	Peer review of features
4	Spot-check harness: ≥ 8 families, 2 baselines	Spot-check report
5	Optuna, calibration, profit threshold; stub API deployed	**D2 (15%)**
6	SageMaker script mode + HPO, artifact + sidecar, endpoint torn down	Cost report
7	MLflow tracking, registry, promotion policy (buffer week)	Registry evidence
8	Association rules, serving engine, hit-rate evaluation	**D3 (15%)**, tag `v0.3`
9	Policy corpus (~15 pp.), index, tools; golden set of 40	Golden set submitted
10	Three providers, guardrails, assistant evaluation	Evaluation table
11	API + containers complete, one-command stack	Cold-start test
12	CI/CD with OIDC, live URL	**D4 (15%)**, tag `v0.9`
13	Five-section interface (buffer week)	Live application link
14	Monitoring + drift, fairness audit, Law 25 file	**D5 (10%)**
15	Report, bilingual executive summary, deck, defence	**D6 (15%) + D7 (20%)**, tag `v1.0`

9. Cloud cost plan and budget control

Ceiling **$25**, target **< $12**: SageMaker HPO on managed spot with 15-minute caps (≈ $0.05–0.50), one real-time endpoint for < 1 h (≈ $0.10), S3 < $0.50, serving on Render's free tier or App Runner paused outside demos, hosted-LLM evaluation on synthetic questions only (< $2). Region `ca-central-1`. Training is local by default; managed training is used where the assignment requires evidence, and its crossover point is analysed in D3. The HPO backend is a configuration switch (`local` / `aws` / `azure`, `docs/CLOUD_OPTIONS.md`): SageMaker is the plan of record; Azure ML (Canada Central, low-priority VMs) is prepared as a fallback to be used only with the instructor's approval. Budget runbook: `docs/runbooks/AWS_BUDGET_ALARM.md`.

> **Control point — budget alarm evidence.** A $10 monthly cost budget with alerts at 50%, 85% and 100% forecast is part of this cost plan. Its configuration must be verified in the AWS console and the screenshot attached to this section before submission (steps in the budget runbook).

10. Sources and AI-tool use

Chen, D. (2012). *Online Retail II* [Dataset]. UCI Machine Learning Repository, https://doi.org/10.24432/C5CG6D (CC BY 4.0). Course specification *Capstone Project 1* (Vanier College, 420-977-VA). AI-assisted coding and drafting tools were used for code, tests and report text; every section was reviewed, and all numbers were checked against the pipeline outputs. Where AI-generated output had to be corrected is recorded in D3 §15.